

I'll be frank, Descarte wasn't the brightest philosopher. Sure, he was the birthplace of "I think therefore I am," a rhetorical thought exercise known by millions, but the path through which he arrives at this answer is convoluted and uninformed. In book two of *Meditations*, Descarte blathers on extensively about a piece of wax, describing in intricate detail the wax and its physical properties and then how the mind understands that when the wax melts, it is the same wax, despite the changes in color or smell or shape. I will argue that had Descarte been taught rudimentary chemistry he would have been able to explain the transformation of the wax without defaulting to "the mind alone," and revealing it through "my imagination." I would hope that a celebrated philosopher would have better proof for something than the justification a child uses when arguing their imaginary friend exists.

Descartes starts by noting all the physical features of the wax, a sign that by this point in his meditations, the mind is trusting the senses (despite the deceiving evil demon.) He talks at length about the many characteristics of the wax, saying "It has just been taken from the honeycomb; it has not yet quite lost the taste of the honey; it retains some of the scent of the flowers from which it was gathered; its color, shape and size are plain to see; it is hard, cold and can be handled without difficulty; if you rap it with your knuckle it makes a sound." It seems like Descarte got to know this piece of wax quite well, though only by observing the wax. I would argue that instead of observing the wax, understanding the reasons for its properties is vastly more important. If, as a child, Descartes was taught simple molecular theory, he would understand that wax is a solid, made of long strands of complex hydrocarbons that are bound together. Within the solid are aromatic hydrocarbons, leading to the color and distinct smell, and that the flexible nature of these molecules allows it to be transformed and molded into different shapes. The key shift that Descarte has to enact is changing from using the physical senses to the mind's sense of understanding. If Descarte would

peer into the wax and see not its smell nor its color nor its taste, he would see with his mind a collection of molecules, differing in size from large and small, each adding its characteristics to the wax as a whole. In addition, if he continued to be taught about the simple chemistry of organic molecules such as those that make up the lump of wax he seems so focused on, he would be able to address the second conundrum he faces: the “immeasurable” changes the wax can go through.

A start moment of Descartes inability to think scientifically comes when he argues that the mind cannot comprehend the complexity of the wax. He says “Not at all; for I can grasp that the wax is capable of countless changes of this kind, yet I am unable to run through this immeasurable number of changes in my imagination, from which it follows that it is not the faculty of imagination that gives me my grasp of the wax as flexible and changeable.” Descartes is arguing that the imagination is weak, and that it cannot be through imagination that we understand the wax as we perceive it.

Descarte is true about one thing, the mind cannot comprehend “infinite”. If I handed you a piece of cosmic energy and told you to explain the infinite ways it could change, our limited minds would be overwhelmed with the possibilities. But for something as simple as wax, the possibilities are whittled down to a reasonable amount, especially when taking into account the source of the change. We can understand through our mind that when a piece of wax is held to a flame, the molecules within are heated, and in their state of high energy are able to move in such a way that causes them to be a liquid. This is a change we can quantify and understand, and as such melting points are a common statistic in determining different solids. Ice melts at a low temperature, while iron melts at higher temperatures. The process is the same, and we understand the process by which something melts. Despite this understanding, I already hear Descarte complaining, telling me that while the mind can understand change in one tiny molecule of wax, the wax is too large to

comprehend all the changes at once. And to this I will tell him, “look towards the forest and tell me what you see.” Unless he aims to be pedantic, he will respond with a description of the forest, and how the leaves sway as the breeze travels through the canopy. It is here that I will explain how a leaf grows, starting from a bud and unfurling over the days to a broad leaf, green and rugged. I will then point to the countless leaves in the forest and ask “So how did all of those leaves grow?” Unless he chooses to be ignorant, he will respond by saying “They each grew from a bud, unfurling over the days.” It is in this way that we may understand the wax. For we can understand the simplicity of a single molecule and its flexibility, and how another molecule gives the wax its color, and yet another its smell. Once we understand the pieces, with our imagination and our mind we can collect them all together and construct a final image, similar to how one sees a forest and does not think about the roots underground or the leaves on the branches or the bark on the wood, although we can understand each individually. Most people simply see a forest, and do not think twice.

Lastly we come to Descartes final conclusion, that “when I distinguish the wax from its outward forms—take the clothes off, as it were, and consider it naked—then although my judgment may still contain errors, at least my perception now requires a human mind.” While his method was wrong, he came to the right conclusion. It is through the mind, and understanding each individual part of the wax, that we are able to perceive the true nature of the wax and follow its changes until it no longer becomes wax (such as when it burns and leaves nothing behind.) He was able to understand that it was through the power of the mind that he could understand the wax past its physical properties. If he had been taught basic molecular theory, he would have argued that it was through the mind’s sense of understanding what cannot be seen that we extend the idea of the wax, and therefore is “perceived by the mind alone.”